

Synthesis Review: Vector Motion, Polar Curves, and Accumulation

Answer Key

AP Calculus BC

Quick Answers

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| 1. the speed = 5.0 | 7. the x-coordinate = 6, the y-coordinate = 4 |
| 2. the intersection angles = $\frac{\pi}{3}, \frac{5\pi}{3}$ | 8. the intersection angles = $\frac{\pi}{3}, \frac{5\pi}{3}$, the radius there = 1.5, — |
| 3. the litres added = 5π L | 9. the litres delivered = $60 - \frac{60}{e}$ L, the half-rate time = 3.466 min |
| 4. $dx/dt = 2 \cos(2t)$, $dy/dx = 1.047$ | 10. the total distance = 12 m, the x-coordinate at $t = 3 = 9$, the y-coordinate at $t = 3 = 6$, — |
| 5. the area = $\frac{4\pi}{3} + 2\sqrt{3}$ | |
| 6. the distance = $10\sqrt{5} - 2$ m | |
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What is verified

16 of 18 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus BC

FUN-3.C – problems 1, 4, 7, 10

FUN-6 – problems 2, 3, 5, 6, 8, 9, 10

Difficulty 1–5 of 5 across 18 tagged checks.

1. $x(t) = t^2 + 1$, $y(t) = 3t$. Find the speed at $t = 2$.

Solution: $x'(t) = 2t$ and $y'(t) = 3$, so at $t = 2$ the velocity is $\langle 4, 3 \rangle$ and the speed is $\sqrt{4^2 + 3^2} = \sqrt{25}$. 5

2. Where do $r = 2$ and $r = 4 \cos \theta$ meet, for θ in $[0, 2\pi)$?

Solution: Set the two expressions for r equal: $4 \cos \theta = 2$, so $\cos \theta = \frac{1}{2}$. On $[0, 2\pi)$ that gives $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$

3. $r(t) = 5 + 2 \cos t$ litres per hour. How much enters between $t = 0$ and $t = \pi$?

Solution: Accumulated amount is the integral of the rate:

$$\int_0^\pi (5 + 2 \cos t) dt = \left[5t + 2 \sin t \right]_0^\pi = 5\pi + 0.$$

So the reservoir gains

$$5\pi \approx 15.708 \text{ L}$$

4. $x = \sin(2t)$, $y = t^2$. (a) Find $\frac{dx}{dt}$. (b) Find $\frac{dy}{dx}$ at $t = \frac{\pi}{6}$.

Solution (a): By the chain rule,

$$\frac{dx}{dt} = 2 \cos(2t)$$

Solution (b): $\frac{dy}{dt} = 2t$, so $\frac{dy}{dx} = \frac{2t}{2 \cos(2t)}$. At $t = \frac{\pi}{6}$ the denominator is $2 \cos \frac{\pi}{3} = 1$ and

the numerator is $\frac{\pi}{3}$.

$$\frac{dy}{dx} \approx 1.047$$

5. Area inside $r = 4 \cos \theta$ and outside $r = 2$.

Solution: The curves meet at $\theta = \pm \frac{\pi}{3}$, and between those angles $4 \cos \theta \geq 2$, so

$$A = \frac{1}{2} \int_{-\pi/3}^{\pi/3} \left((4 \cos \theta)^2 - 2^2 \right) d\theta = \left[4\theta + 4 \sin 2\theta \right]_0^{\pi/3}.$$

Evaluating gives

$$A = \frac{4\pi}{3} + 2\sqrt{3} \approx 7.653$$

6. $x(t) = 3t$, $y(t) = 2t^{3/2}$ on $[0, 4]$, metres. Find the distance travelled.

Solution: $x'(t) = 3$ and $y'(t) = 3\sqrt{t}$, so the speed is $\sqrt{9 + 9t} = 3\sqrt{1 + t}$ and

$$\int_0^4 3\sqrt{1+t} dt = 2 \left[(1+t)^{3/2} \right]_0^4 = 2(5\sqrt{5} - 1).$$

So the particle travels

$$10\sqrt{5} - 2 \approx 20.361 \text{ m}$$

7. $x'(t) = 4 \cos t$, $y'(t) = 4 \sin t$, starting at $(2, 0)$. Find the position at $t = \frac{\pi}{2}$.

Solution: Integrate each component from its starting value: $x(t) = 2 + 4 \sin t$ and $y(t) = 4 - 4 \cos t$. At $t = \frac{\pi}{2}$, $\sin = 1$ and $\cos = 0$, so the coordinates are $2 + 4$ and $4 - 0$.

$$(x, y) = (6, 4)$$

8. $r = 1 + \cos \theta$ and $r = 3 \cos \theta$. (a) Solve simultaneously. (b) Find r there. (c) Why does the algebra miss the pole?

Solution (a): $1 + \cos \theta = 3 \cos \theta$ gives $\cos \theta = \frac{1}{2}$, so on $[0, 2\pi)$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Solution (b): At either angle, $r = 3 \cos \frac{\pi}{3} = 3 \left(\frac{1}{2} \right)$.

$$r = 1.5$$

Solution (c) — instructor-judged. Both curves do pass through the pole, but at different parameter values: the cardioid reaches $r = 0$ at $\theta = \pi$, and the circle reaches $r = 0$ at $\theta = \frac{\pi}{2}$. Since no single θ makes both radii zero, the simultaneous equations cannot produce it. A polar point has many (r, θ) names, so the pole must always be checked separately. Full credit says both of those things.

9. $r(t) = 12e^{-0.2t}$ litres per minute. (a) Amount in the first 5 minutes. (b) When is the rate half its initial value?

Solution (a):

$$\int_0^5 12e^{-0.2t} dt = \left[-60e^{-0.2t} \right]_0^5 = 60 - 60e^{-1}.$$

So the pipe delivers

$$60(1 - e^{-1}) \approx 37.93 \text{ L}$$

Solution (b): The initial rate is $r(0) = 12$, so solve $12e^{-0.2t} = 6$, i.e. $e^{-0.2t} = \frac{1}{2}$ and $-0.2t = -\ln 2$.

$$t = 5 \ln 2 \approx 3.466 \text{ min}$$

10. Velocity $\langle 2t, t^2 - 1 \rangle$ m/s from the origin, $0 \leq t \leq 3$. (a) Distance. (b) Position at $t = 3$. (c) Why is (a) larger?

Solution (a): The speed simplifies exactly:

$$\sqrt{(2t)^2 + (t^2 - 1)^2} = \sqrt{t^4 + 2t^2 + 1} = t^2 + 1,$$

so $\int_0^3 (t^2 + 1) dt = \left[\frac{t^3}{3} + t \right]_0^3 = 9 + 3$. The distance is

$$12 \text{ m}$$

Solution (b): Starting from the origin, $x(t) = t^2$ and $y(t) = \frac{t^3}{3} - t$, so at $t = 3$ the coordinates are 9 and $9 - 3$.

$$(x, y) = (9, 6)$$

Solution (c) — instructor-judged. The straight-line distance from the origin to $(9, 6)$ is $\sqrt{117} \approx 10.817$, less than 12. The path is curved: integrating the speed adds up every centimetre actually travelled, while the displacement compares only the two endpoints. The two agree only when the motion is along a straight line in a single direction. Full credit makes that comparison.